Computational Physics
Project III: The carbon cycle and climate response

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1 Introduction

The human induced emission of greenhouse gases, such as carbon dioxide and methane, has increased the average global temperature with more than 1 °C since pre-industrial times. This is due to the increased absorption of thermal radiation by greenhouse gases in the atmosphere, which in turn decreases the amount of thermal radiation escaping to space. To restore the energy balance, the earth's surface and lower atmosphere will warm up. In this project, your task is to model previous and future carbon dioxide emissions using a box model and investigate how assumed future emission scenarios affect the evolution of the climate in terms of global temperature.

2 Theory – Carbon cycle box model

In the project you will study a box model for the carbon cycle which was suggested by Akira Tomizuka [1]. The theory of the box model is summarized below, but for a complete description we encourage you to read the original paper, which is available on Canvas. The box model is depicted in figure 1 and consists of seven compartments. The value given in each of the compartments is the preindustrial content of carbon in units of PgC, where 1 PgC = 1×10^{15} g (gram) of carbon. Using the molar weights of carbon (C) and carbon dioxide (CO_2) we find that 1 g of carbon equals approximately 3.66 g of carbon dioxide. This means that the total content of carbon dioxide in the atmosphere is 615 PgC $\cdot$ 3.66 = 2251 Gt. To get the atmospheric concentration of CO_2 in ppm (parts per million) we divide the content in PgC with 2.13. Thus, we realize that 615 PgC corresponds to $615/2.13 \approx 289$ ppm, which is thus the preindustrial concentration. The rate equation for each compartment is given as

$$\begin{aligned}
\frac{dN_1}{dt} &= -k_{12}N_1 + k_{21}\left(N_2^0 + \xi\left(N_2 - N_2^0\right)\right) + \gamma - f + \delta + k_{51}N_5 \\
&\quad + k_{71}N_7\,, && (1)\\
\frac{dN_2}{dt} &= k_{12}N_1 - k_{21}\left(N_2^0 + \xi\left(N_2 - N_2^0\right)\right) - k_{23}N_2 + k_{32}N_3 - k_{24}N_2\,, && (2)\\
\frac{dN_3}{dt} &= k_{23}N_2 - k_{32}N_3 - k_{34}N_3 + k_{43}N_4\,, && (3)\\
\frac{dN_4}{dt} &= k_{34}N_3 - k_{43}N_4 + k_{24}N_2 - k_{45}N_4\,, && (4)\\
\frac{dN_5}{dt} &= k_{45}N_4 - k_{51}N_5\,, && (5)\\
\frac{dN_6}{dt} &= f - k_{67}N_6 - 2\delta\,, && (6)\\
\frac{dN_7}{dt} &= k_{67}N_6 - k_{71}N_7 + \delta\,. && (7)
\end{aligned}$$

The preindustrial values of the transfer coefficients k_{ij} in y^{-1} (per year) are given in table 1. For example, the transfer coefficient from compartment 1 (atmosphere) to compartment 2 (surface ocean) is denoted k_{12} and has the

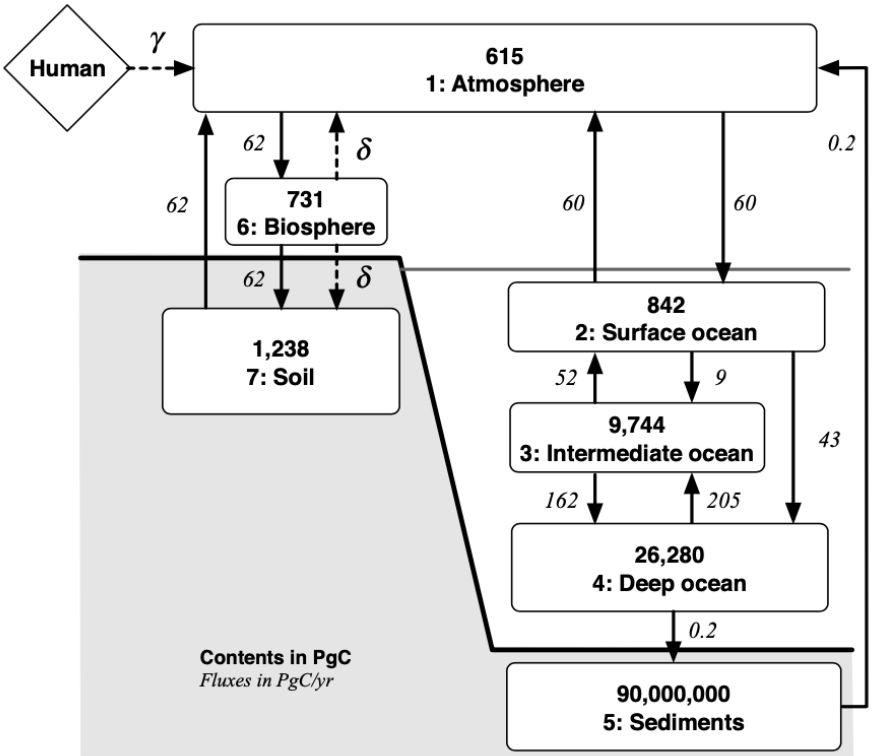

Figure 1: *The seven-box model for the global carbon cycle. The assumed values of the content and flux in the preindustrial era are expressed in PgC and PgC/yr, respectively. Figure taken from [1].*

preindustrial value $60/615 \approx 0.098$ y^{-1}. The term $k_{21}\left(N_2^0 + \xi\left(N_2 - N_2^0\right)\right)$ in equations 1 and 2 accounts for the fact that the CO_2 uptake by the ocean is suppressed due to the large amount of dissolved inorganic carbon in the ocean. Here ξ is the unitless buffer factor which depends on P, i.e., the atmospheric

Table 1: Values of preindustrial transfer coefficients k_{ij} in units of y^{-1} and other constants mentioned in the text.

$k_{12} = 60/615$		
$k_{21} = 60/842$	$k_{23} = 9/842$	$k_{24} = 43/842$
$k_{32} = 52/9\ 744$	$k_{34} = 162/9\ 744$	
$k_{43} = 205/26\ 280$	$k_{45} = 0.2/26\ 280$	
$k_{51} = 0.2/90\ 000\ 000$		
$k_{67} = 62/731$		
$k_{71} = 62/1\ 238$		
$f_0 = 62$ PgC/year		
$\beta = 0.42$		
$P_0 = 289$ ppm		
$N_2^0 = 842$ PgC		

CO_2 concentration in ppm, according to the following parametrization

$$\xi(P) = 3.69 + 1.86 \cdot 10^{-2} P - 1.80 \cdot 10^{-6} P^2 \,, \tag{8}$$

and N_2^0 is the preindustrial content of carbon in the sea surface with the value 842 PgC. We note that when $\xi = 1$, the term reduces to $k_{21} N_2$ (no buffer effect). Further on, γ and δ are carbon emissions in PgC/year due to fossil fuel burning (γ) and land usage (δ), respectively. Historical emission data, in units of metric tonnes CO_2 per year (t/y), can be downloaded from [2]. Finally, f, is the net CO_2 flux to the biosphere (the difference between the carbon uptake rate by photosynthesis and the carbon emission rate by respiration) and is given as

$$f = f_0 \left(1 + \beta \ln \left(\frac{P}{P_0} \right) \right) \,, \tag{9}$$

where $f_0 = 62$ PgC/year is the preindustrial flux, P_0 and P are the preindustrial and current atmospheric concentrations of CO_2, respectively, and β is the so-called fertilization factor, which will be treated as the only free parameter and used to match the present observed atmospheric CO_2 concentration (it's default value is usually around 0.42). Solving the equation system will give $N_i(t)$ and in this project we are especially interested in $N_1(t)$, which is the content of carbon in the atmosphere as a function of time expressed in units of PgC. Dividing $N_1(t)$ with 2.13 yields $P(t)$, the concentration of atmospheric CO_2 in ppm.
In figure 2 CO_2 emission input data and resulting atmospheric CO_2 concentration for years 1750 – 2020 is seen. In the simulation $\beta = 0.42$ was used.

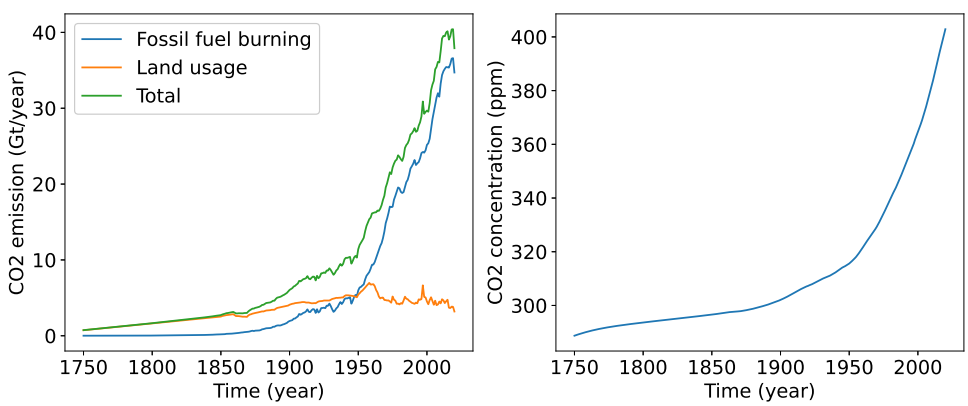

Figure 2: *Input CO_2 emissions from [2] (left) and resulting atmospheric CO_2 concentration (right) for years 1750 – 2020 assuming $\beta = 0.42$. Note that input data for land usage is only available from year 1850, and in the figure CO_2 emission due to land usage is assumed to be zero in year 1750 and growing linearly to year 1850.*

3 Theory – Climate sensitivity

As described in the introduction, increased concentrations of greenhouse gases in the atmosphere decreases the amount of thermal radiation from the earth/atmosphere that escapes to space. The difference in intensity between the incoming unreflected solar radiation and the outgoing thermal radiation is called radiative forcing, in short RF, with symbol ΔF and unit $\mathrm{W/m^2}$. RF is thus a measure of the energy imbalance of the earth/atmosphere system and $\Delta F > 0$ implies that the earth is getting warmer. For CO_2 we have the following expression

$$\Delta F = 5.35 \cdot \ln\left(\frac{P}{P_0}\right) . \tag{10}$$

If we know ΔF, the resulting temperature increase ΔT of the earth/atmosphere can be computed as

$$\Delta T = \lambda \Delta F \,, \tag{11}$$

where λ is the so-called climate sensitivity parameter with unit $\mathrm{Km^2/W}$. In the absence of feedback mechanisms, we find that for CO_2 $\lambda = 0.27\ \mathrm{Km^2/W}$. However, we know that feedback mechanisms exist and are important. As an example, a consequence of a warmer atmosphere is that the atmosphere on average will contain more water vapor, and since water vapor is a greenhouse gas, this will warm the atmosphere even more, and so on. This would be a positive feedback mechanism. There exist also negative feedbacks, for example a warmer earth could trigger forest fires which will release small particles, aerosols, in the atmosphere which on average will have a cooling effect since more of the incoming solar radiation will be reflected. However, most feedback mechanisms are positive, and we expect $\lambda > 0.27\ \mathrm{Km^2/W}$, however the exact value of λ is uncertain and subject to scientific interest.

In this project, we will estimate the climate sensitivity based on observations of atmospheric CO_2 concentration and global temperature anomaly as a function of time, which is exemplified in figure 3. By using equations 10, 11 and

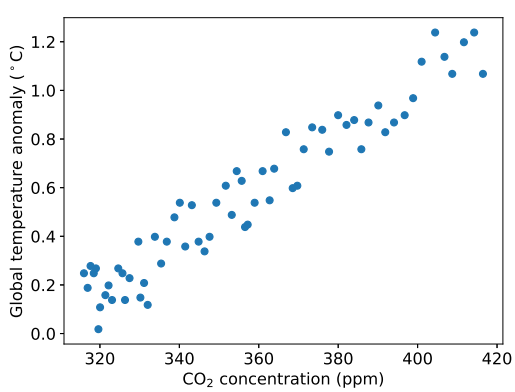

Figure 3: *Global mean temperature anomaly from NASA/GISS [3] as a function of CO_2 concentration as recorded on Mauna Loa [4] for years 1959-2021.*

data displayed in figure 3, it is straight forward to quantify the relation between temperature and atmospheric CO_2. Note, however, that the increase in temperature is not only due to emission of CO_2. On the other hand, the contribution from CO_2-emissions is by far the largest and if we implicitly assume, when modeling future CO_2 emissions, that remaining contributors behave as CO_2, then it is justified to use data as in figure 3.

4 Project Task

1. Find and download suitable historic CO_2 emission data. Data must be separated in fossil fuel burning and land usage and, if possible, extend back to 1750.

2. Using the box model described and downloaded data, solve the system of differential equations assuming $\beta = 0.42$.

3. Modify the value of β and run the box model again. Repeat until the simulated CO_2 concentration agrees with the observed current concentration.

4. Implement different future scenarios for CO_2 emissions until 2100, run the box model and record how the global temperature develops. For emission scenarios, see for example page 13 in [5] for inspiration.

5. Compare your results with other sources.

References

[1] A. Tomizuka, American Journal of Physics 77, 156 (2009)

[2] `https://ourworldindata.org/co2-emissions`

[3] `https://data.giss.nasa.gov/gistemp/graphs_v4/`

[4] `https://gml.noaa.gov/ccgg/trends/data.html`

[5] `https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_SPM.pdf`